Recipe 2: Managing Variability

Step 1: Write down all the parameters λ, τ, s, CV_a, CV_s

Customer parameters

- Arrival rate, λ: number of customers *arriving* per unit time (e.g., $10\,[{}^{customers}/_{h}]$)
- Average interarrival time (IAT): average time between two consecutive arrivals, $\overline{IAT} = \frac{1}{\lambda}$
- Coefficient of variation for arrivals, CV_a: extent of variability of arrivals in relation to their mean,
$$CV_a = \frac{\text{standard deviation of IAT}}{\text{mean of IAT}}$$
Example: Poisson arrivals of mean λ have exponentially distributed interrarival times with std.dev. = mean = $\frac{1}{\lambda}$, hence $CV_a = \frac{1/\lambda}{1/\lambda} = 1$.

Service parameters

- Service rate (capacity), μ: number of customers *served* per unit time (e.g., $20\,[{}^{customers}/_{h}]$)
- Average service time, τ: average time it takes to serve one customer, $\tau = \frac{1}{\mu}$ $(0.05\,[h])$
- Coefficient of variation for service, CV_s: extent of variability of services offered in relation to their mean,
$$CV_s = \frac{\text{standard deviation of service time}}{\text{average service time}} = \frac{\text{standard deviation of service time}}{\tau}$$
Example: Service times that are distributed exponentially with parameter μ: std.dev. = mean = $\frac{1}{\mu}$, thus here $CV_s = \frac{1/\mu}{1/\mu} = 1$.

Performance measures

- L_q: average number of customers waiting in the queue
- L: average number of customers waiting in the system overall (queued + on service)
- W_q: average waiting time in the queue
- W: average total waiting time in the system (queue + service), $W = W_q + \tau$
- **Little's Law**: $L_q = \lambda \cdot W_q$, and $L = \lambda \cdot W$

Step 2: Find the number of servers s

Step 3: Find the utilization of a server ρ

$$\rho = \frac{\lambda}{\mu \cdot s} = \frac{\lambda \cdot \tau}{s}$$

Step 4: Calculate the average waiting time in the queue

Calculations here depend on the number of servers s:

- Case 1: If $s = 1$, use the following formula for W_q:

$$W_q = \tau \frac{\rho}{1-\rho} \frac{1}{2} \left(CV_a^2 + CV_s^2\right)$$

- Case 2: If $s > 1$ (there are multiple servers), you can either (1) find L_q from the table at the end of the review sheet for your ρ.[1] After, you can use Little's law to calculate $W_q = \frac{L_q}{\lambda}$; or (2) use the waiting time formula for multiple servers

$$W_q = \frac{\tau}{s} \frac{1}{1-\rho} \frac{1}{2} \left(CV_a^2 + CV_s^2\right) \rho^{(\sqrt{2\,(s+1)}-1)}$$

Step 5: Calculate the average total waiting time in the system

Average total waiting time in the system = queueing time + service time:

$$W = W_q + \tau$$

Step 6: Calculate the average number of customers

If you haven't yet found L_q in step 4, then calculate it using Little's law:

$$L_q = \lambda \cdot W_q$$

For the total number of customers in the system you can also use Little's law:

$$L = \lambda \cdot W$$

[1] Note that, sometimes, the exact value of ρ for which you would like to find L_q is not in the table. In this case, you need to use a linear approximation. First, find the closest value to ρ from below. Denote it by $\underline{\rho}$ and find the corresponding $\underline{L}_q$. Secondly, find the closest value to ρ from above. Denote it by $\overline{\rho}$ and find the corresponding $\overline{L}_q$. Then, the value of ρ for $\underline{\rho} < \rho < \overline{\rho}$ is approximated in the following way:

$$L_q(\rho) = \underline{L}_q + \frac{\overline{L}_q - \underline{L}_q}{\overline{\rho} - \underline{\rho}} \cdot (\rho - \underline{\rho})$$